RESM3002: Advanced Social Data Science

Assignment

ID: 34074759

Word Count: 3199

1. Introduction

Stack Exchange is a network of community-led Q&A sites based on topics. Questions mis-routed to unrelated sites can waste multiple parties' time; either getting ignored, migrated or closed (Stack Exchange Inc., 2026b). An automated classifier for site suggestion can be implemented for users in their question-submission flow to notify them of the appropriate site before posting.

The data for this study consists of four CSVs each relating to a site (English Language (ELU), English Learners (ELR), World Building, Sci-Fi). Each class contains 2,000 questions pulled from the Stack Exchange Data Explorer (SEDE) (Stack Exchange Inc., 2026a). Questions were filtered to body text exceeding 600 characters.

The sites explored contain noticeable overlap, particularly ELU with ELR which diverge from a single original site. World Building and Sci-Fi overlap fictional universe references but differ the discussion of these references.

The report will firstly discuss privacy and ethical considerations, then discuss exploratory data analysis (EDA), followed by the modelling approaches and evaluation of findings. Lastly, AI's contributions for this study will be clearly stated.

1.1 Research Question

Can a Stack Exchange question's category be predicted from its engagement metrics, body, and title to provide effective accuracy for routing to the appropriate site?

2. Privacy and Ethical Considerations

2.1 Class of Information

In SEDE, the data extracted come with user IDs, display names, account creation, location, and links to external accounts (Stack Exchange, 2026b). While display names are usually pseudonyms, the Stack Exchange profile pages are indexed by search engine and public, thus this table would identify a non-negligible proportion of users.

The combination of interests across topics, timestamps, external link fields and writing style can result in a re-identification surface that can be more telling than a standard numeric ID, this is identified as a mosaic effect problem (Solove, 2006; Sweeney, 2002).

2.2 User Privacy Harm

2.2.1 Group Identification

ELR users are majority second-language learners, therefore a classifier that collects users' previous question history can collect proxy signals for ethnicity, migration status, and nationality. Thus, decisions made on these proxies can cause systematically different outcomes for these users, known as the disparate impact failure mode of algorithmic decision systems (Barocas & Selbst, 2016); this is part of the core discriminatory effects identified by Mittelstadt et al. (2016) in the ethics-of-algorithms framework.

2.2.2 Re-identification Through Writing Style

Narayanan et al. (2012) discuss stylometric attributes of anonymous online text as a substantial privacy threat. With this study using thousands of training questions, it can be enough to identify regular contributors within topics with statistically significant accuracy.

2.2.3 Function Creep

A model outputting site suggestion can easily become a model that predicts if a question is asked by a non-native speaker. This can result in prediction not requested by users that also breach Stack Exchange's terms of use (Boyd & Crawford, 2012; Crawford & Paglen, 2019; Stack Exchange Inc., 2026a). This can occur via a model's trained weights applied for a different purpose without a consent procedure, which directly breaches GDPR's purpose limitation principle (Council of the EU, 2016, Art. 5(1)(b)).

2.2.4 Ethical Principles

- Data minimisation
    - Art. 5(1)(c) states if a question text can contribute to a usable classifier, then user attributes are considered excess processing.
- Purpose limitation
    - If a user posted a question for the sole purpose of getting an answer, and did not consider training classifiers, using their data for ML training necessitates a new lawful basis.
- Informed Consent
    - While Stack Exchange content is dual-licensed (CC-BY-SA), users do not knowingly consent to ML model training, therefore CC-BY-SA cannot act as a substitute for ethical consent in categories like nationality inference (Williams et al., 2017; Townsend & Wallace, 2016),
- Algorithmic transparency
    - Classifiers must ensure clear redress mechanisms, as dependence on user metadata can make these mechanisms harder to implement because the cause of error can convolute with user profile data instead of question text, carrying social cost (Diakopoulos, 2016; Mittelstadt et al., 2016).
- Contextual Integrity
    - Conflating user consent to public Q&A data storage with ML-training data consent is a significant breach of big-data ethics (Nissenbaum, 2009)

2.2.5 Constraints in Practice

Considering these ethical implications, the analysis is limited to features derived only from question data (HTML structure, length, body, title and the engagement metrics discussed in (1. Introduction). User-level data is excluded.

3. EDA

3.1 Class Balance

All four classes have 2,000 questions each, thus accuracy is a meaningful metric and applying SMOTE (synthetic minority over-sampling technique (Chawla et al., 2002)) is redundant (Hvitfeldt & Silge, 2021).

World Building has the highest mean answers (4.41) and the lowest unanswered rate (1%), along with the highest mean views (1283), this implies a highly engaged community. The full descriptive statistics table can be found in the (EDA Appendix).

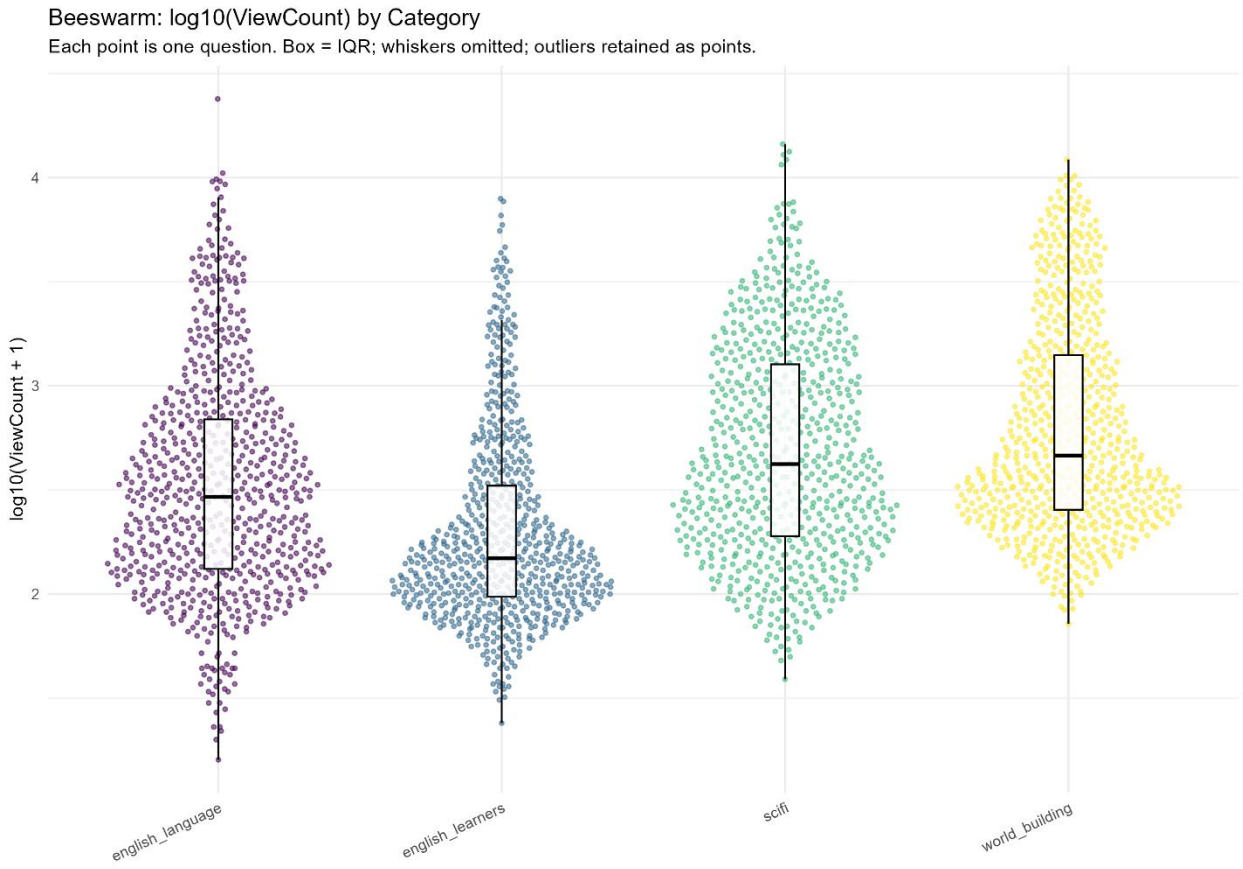


Figure 1

This is supported by a visually tighter spread of view counts, suggesting posts in world building are viewed more on average.

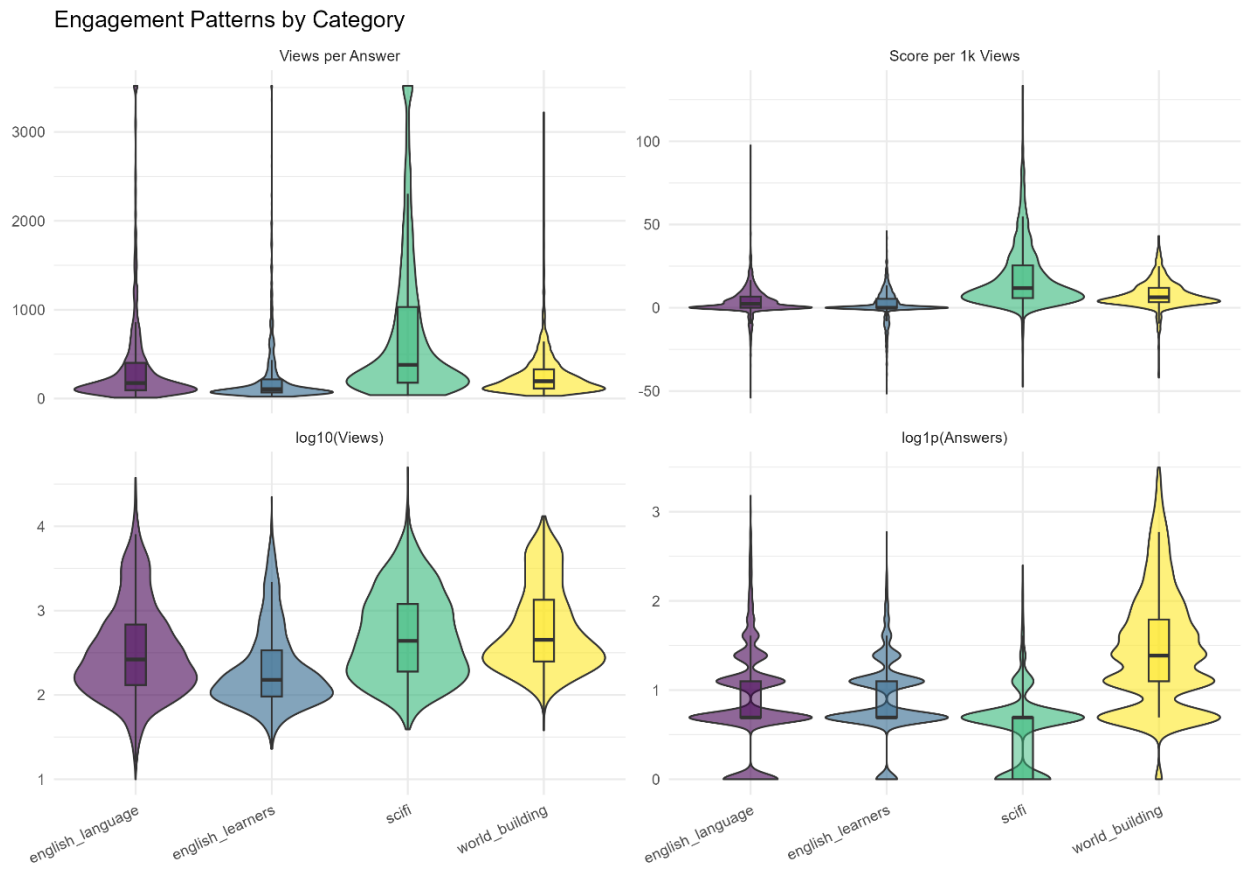


Figure 2

Conversely, when log-transforming answers, world building visually has a greater spread of answers than all other classes.

Sci-Fi contains the highest unanswered rate at 26.5%, this reflects lower perceived relevance of niche fan-trivia questions, this supported by findings that show questions usefulness influence response likelihood (Anderson et al., 2012).

ELU shows greater variance in views when looking at its log-transformed boxplot.

3.2 Complexity and Length

World Building mean word count (275) is the highest among the classes, followed by Sci-Fi (193), ELU (167), and ELR (141), World Building's mean question length is 42.5% longer than the second highest.

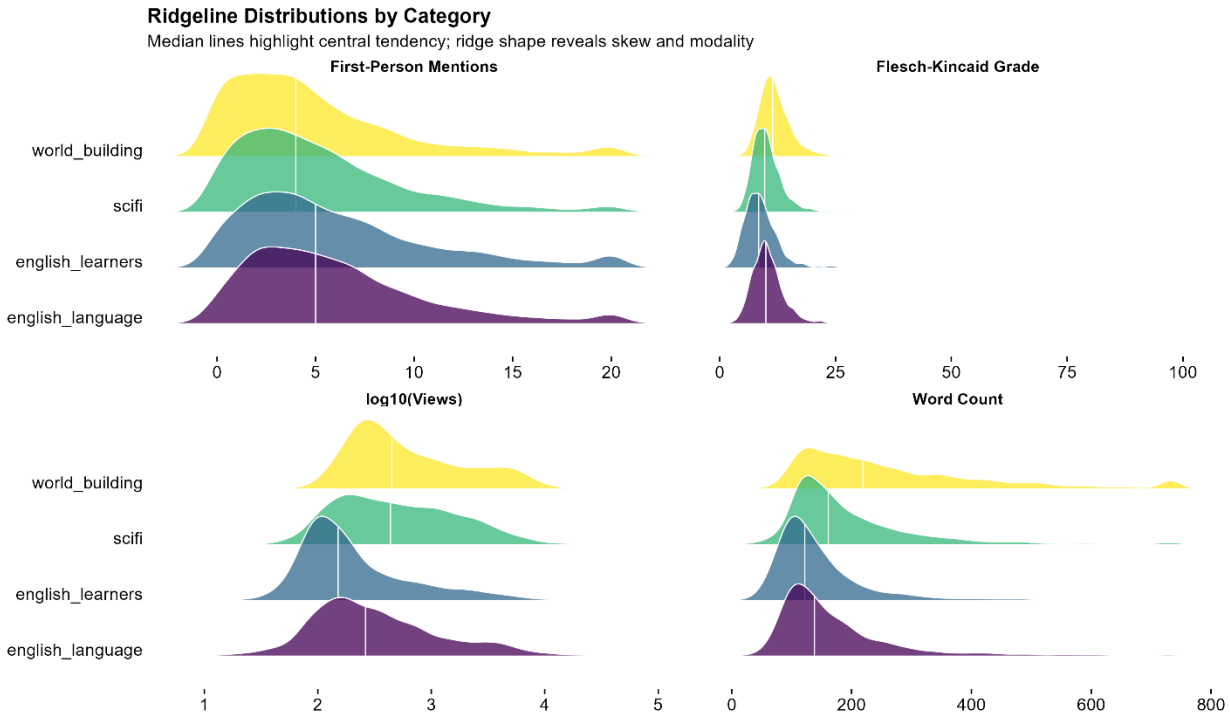


Figure 3

ELU's Flesch-Kincaid grade is 10.4, with ELR's being 8.9. This gap is consistent with the contexts of these sites as ELR caters to learners who are not yet familiar with advanced vocabulary, whereas ELU's members tend to communicate in more specialised terminology (Flesch, 1948).

Table 1

Category	Metric	Mean	Median	S.D
english_language	Dale-Chall	26.90	27.67	9.27
english_learners	Dale-Chall	31.63	32.86	9.78
scifi	Dale-Chall	28.60	29.49	9.52
world_building	Dale-Chall	23.24	24.05	9.64
english_language	Flesch-Kincaid	10.37	9.97	3.51
english_learners	Flesch-Kincaid	8.89	8.41	3.77
scifi	Flesch-Kincaid	10.33	9.72	4.33
world_building	Flesch-Kincaid	12.09	11.44	4.52

english_language	FOG	13.52	13.09	3.80
english_learners	FOG	11.79	11.31	4.03
scifi	FOG	13.22	12.58	4.52
world_building	FOG	15.21	14.55	4.75
english_language	MATTR	0.85	0.86	0.05
english_learners	MATTR	0.83	0.84	0.06
scifi	MATTR	0.87	0.87	0.03
world_building	MATTR	0.88	0.88	0.03
english_language	TTR	0.58	0.58	0.09
english_learners	TTR	0.55	0.56	0.10
scifi	TTR	0.59	0.59	0.07
world_building	TTR	0.58	0.58	0.08

A similar trend can be observed when looking at TTR and MATTR (Covington & McFall, 2010; McCarthy & Jarvis, 2010). ELU's vocabulary is shown to be more diverse on average than ELR. World Building and Sci-Fi are approximately tied with ELU in diversity according to TTR, though ELU marginally lags in MATTR.

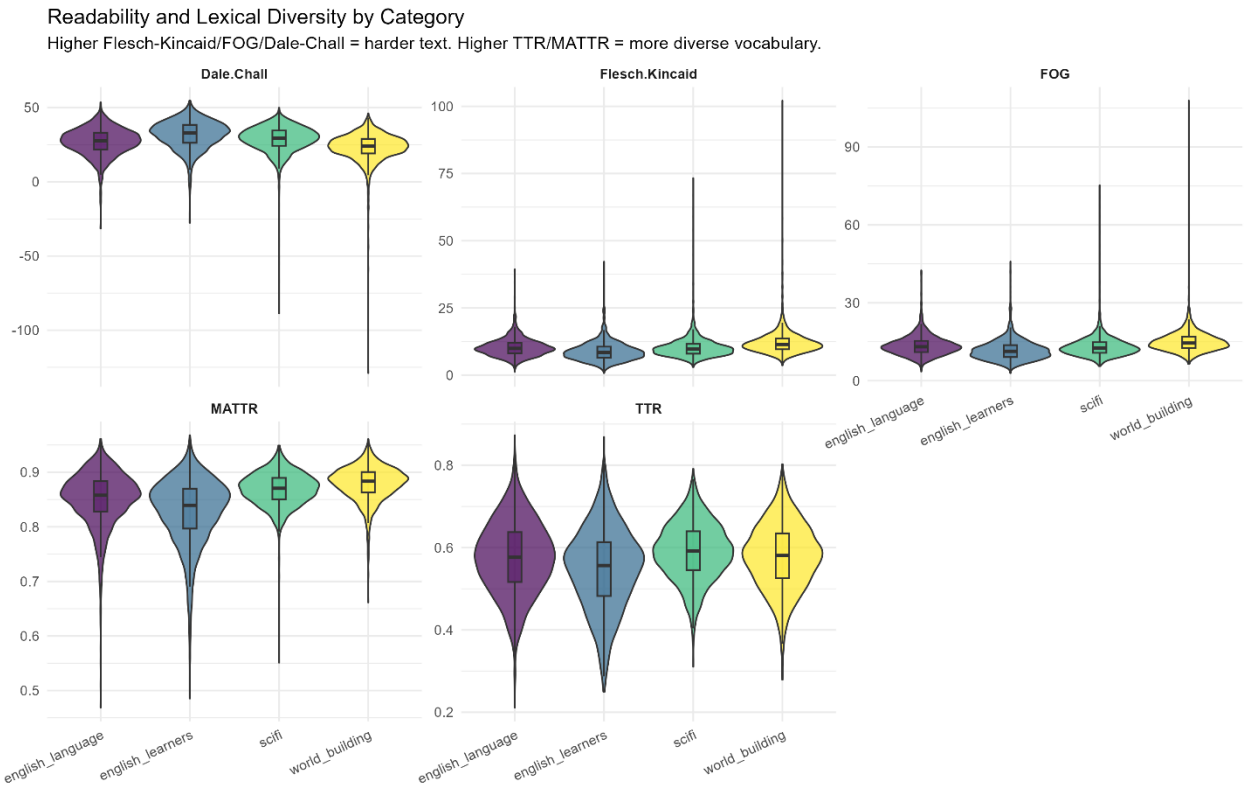


Figure 4

According to FOG and Flesch-Kincaid, World Building has noticeably more difficult text, aligning with the highest average question length.

An outlier to note is the Dale-Chall statistic, showing a reversed order of difficulty, with ELR at the top. This is due to the differences in how Dale-Chall measures difficulty compared to Flesch-Kincaid. The former is sensitive to words outside a list of familiar words observed within a class, this aligns with learners using specific vocabulary and quoted phrases referenced from movies and pop culture, along with foreign-language influenced words. Conversely, world-building appears to share many common words despite being more conceptually complex as Flesch-Kincaid measures difficulty on longer sentences and more syllabic words.

3.3 Lexical Fingerprint

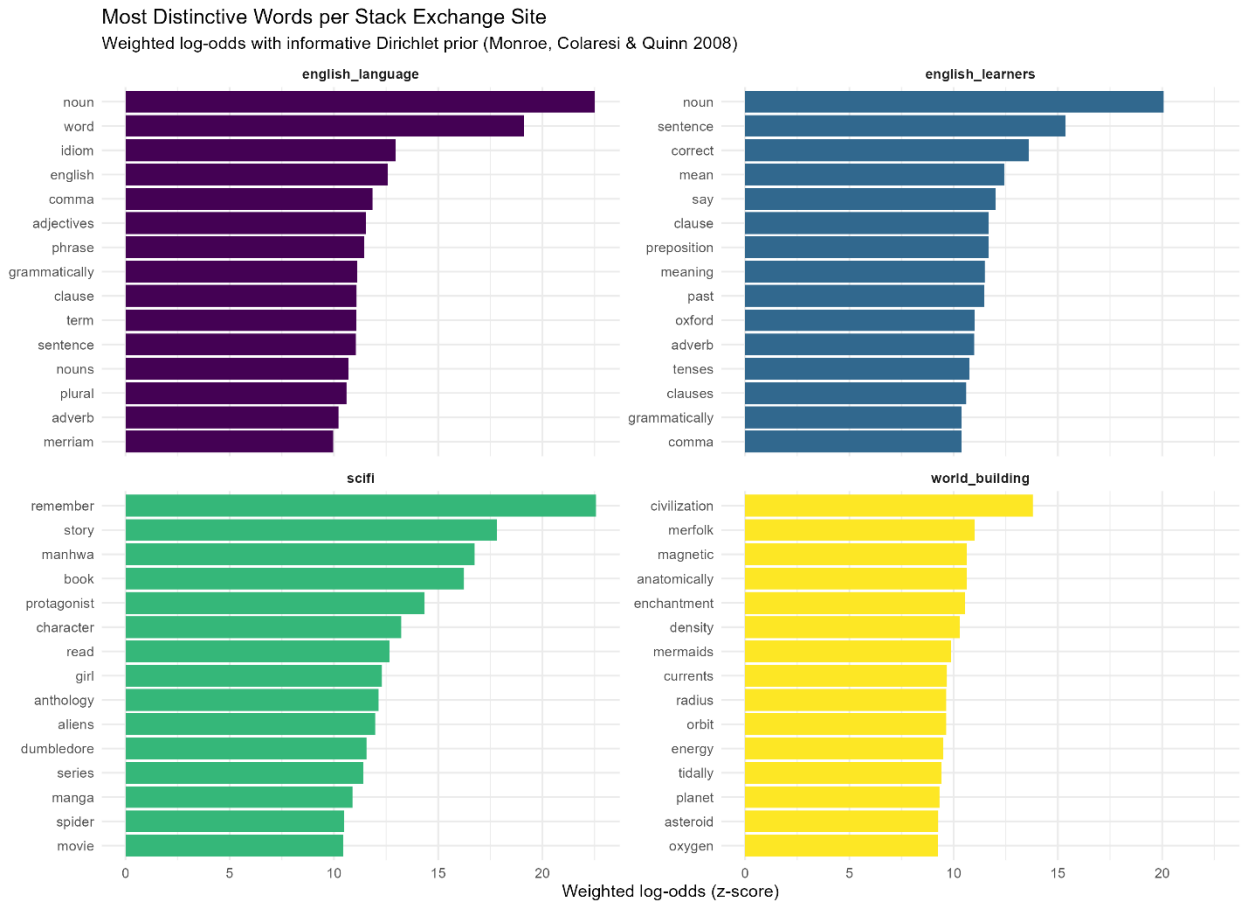


Figure 5

The use of raw TF-IDF (Text Frequency – Inverse Document Frequency) can be dominated by words like ‘the’, ‘is’, etc. Thus, weighted log-odds (Monroe, Colaresi & Quinn, 2008) of the ratio of within-group word to outside-group word probabilities can amend this.

ELR and ELU share ‘noun’ as the highest z-score (~20 and ~25 respectively), along with ‘clause’, ‘comma’, ‘grammatically’, and ‘adverb’. This underlines the evidence of confusion a classifier can face when differentiating between these two classes, however, the association of these differs, as ELU treats them as objects of analysis, and ELR treating them as objects of correctness.

The presence of ‘merriam’ in ELU confirms the words plotted are for etymological answers, and the use of ‘oxford’ and ‘tenses’ confirms a desire to be corrected for users in ELR.

When observing Sci-Fi, the presence of ‘remember’ and ‘dumbledore’ are consistent with the fan-trivia nature of the class, whereas World Building’s presence of ‘anatomically’, ‘asteroid’, and ‘civilisation’ highlight the speculative biological, physical, and anthropological nature of the class. A table of exact log-odds is in the (EDA Appendix).

3.4 Topical Structure

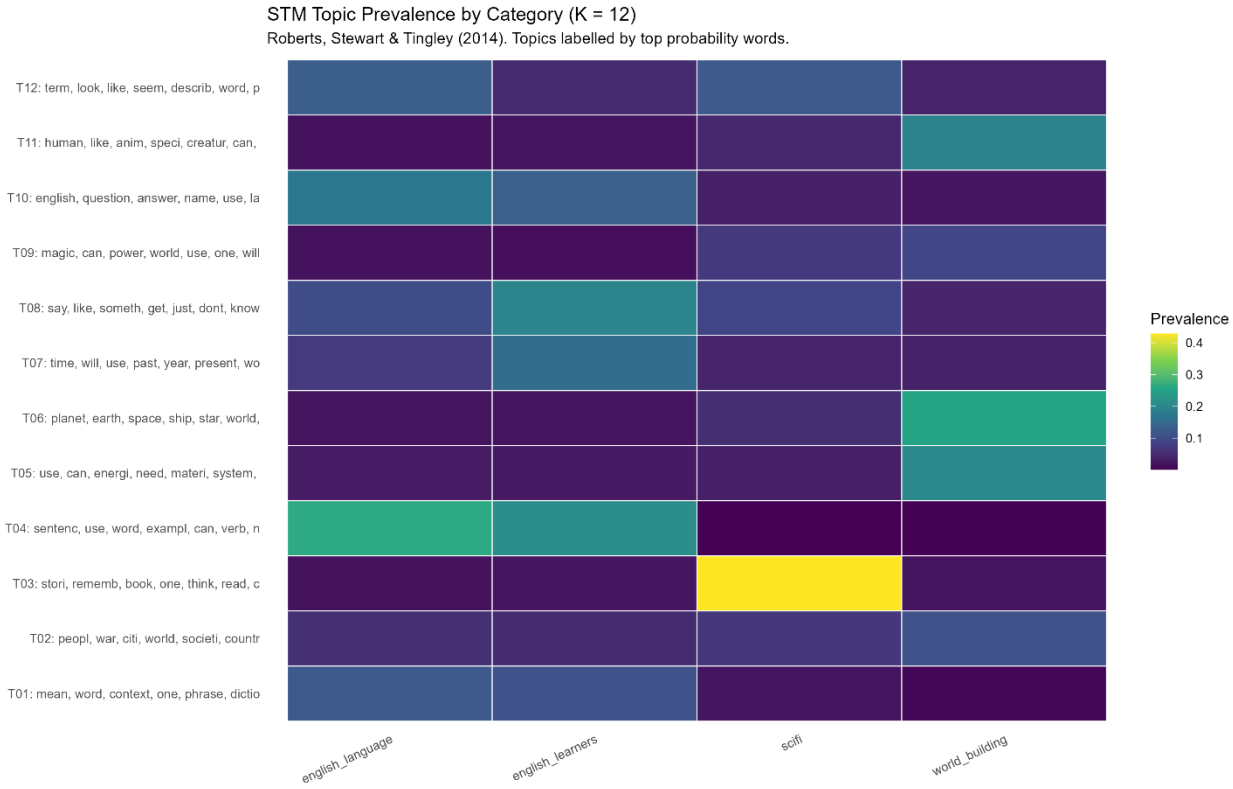


Figure 6

This plot was constructed by a structural topic model, fitting K=12 and prevalence referring to category, generating 12 topics, showing their frequency in each class (Roberts, Stewart & Tingley, 2014).

The frequency in T3 confirms the rationale for Sci-Fi in (3.3 Lexical Fingerprint), having maximum prevalence where all other classes are low. World Building shows the highest prevalence in T6, also supporting its rationale in (3.3 Lexical Fingerprint).

ELU shows higher prevalence in T4, and ELR higher in T8, supporting the rationale discussed in (3.3 Lexical Fingerprint.

3.5 Class Separability

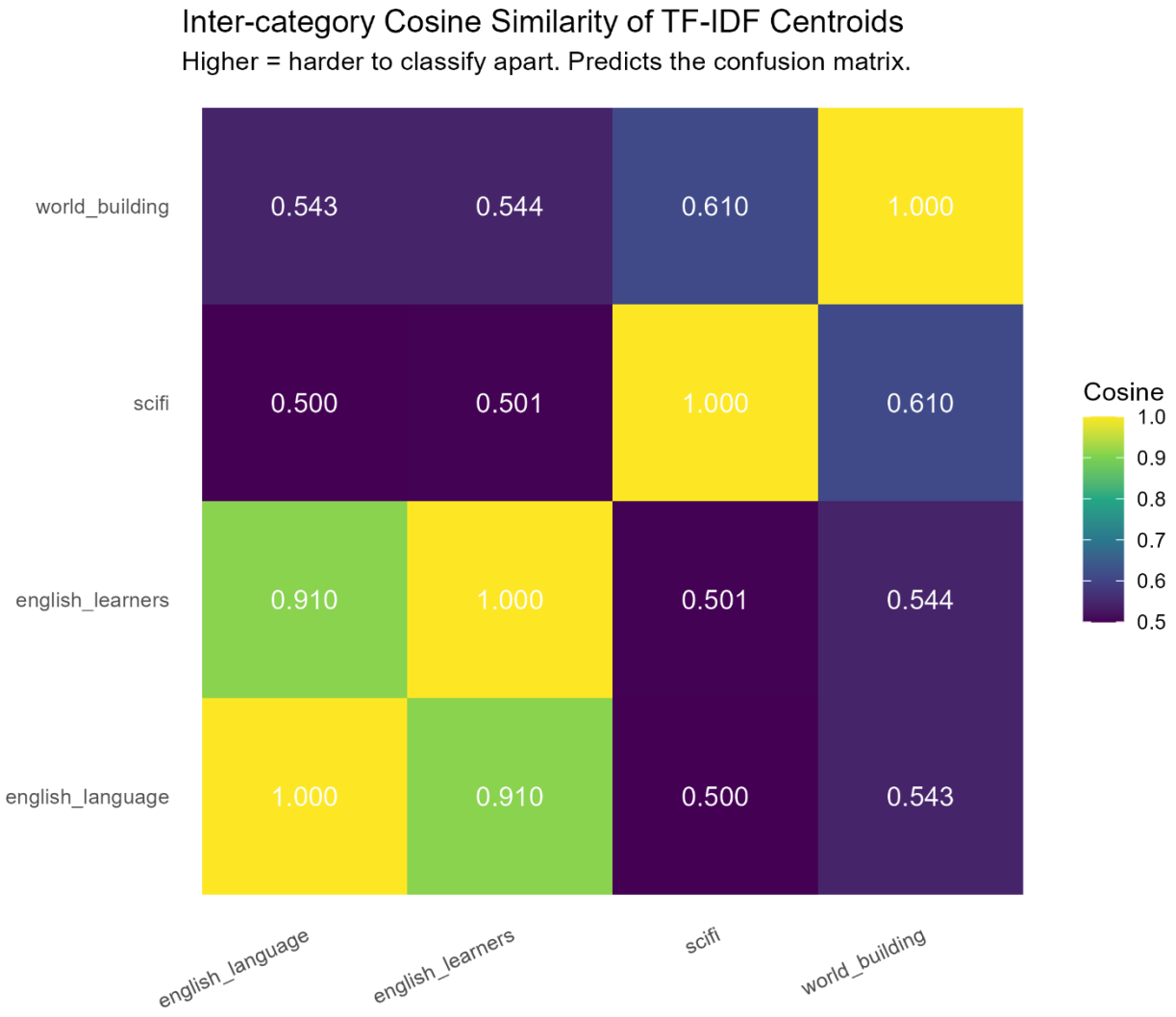


Figure 7

Using TF-IDF, the centroid cosine between ELR and ELU is 0.91, proving their substantial overlap and difficulty in distinguishability for classifiers, while Sci-Fi and World Building share the second highest at 0.61, suggesting classifiers will struggle less differentiating between these 2. All other pairs sit between [0.5, 0.544].

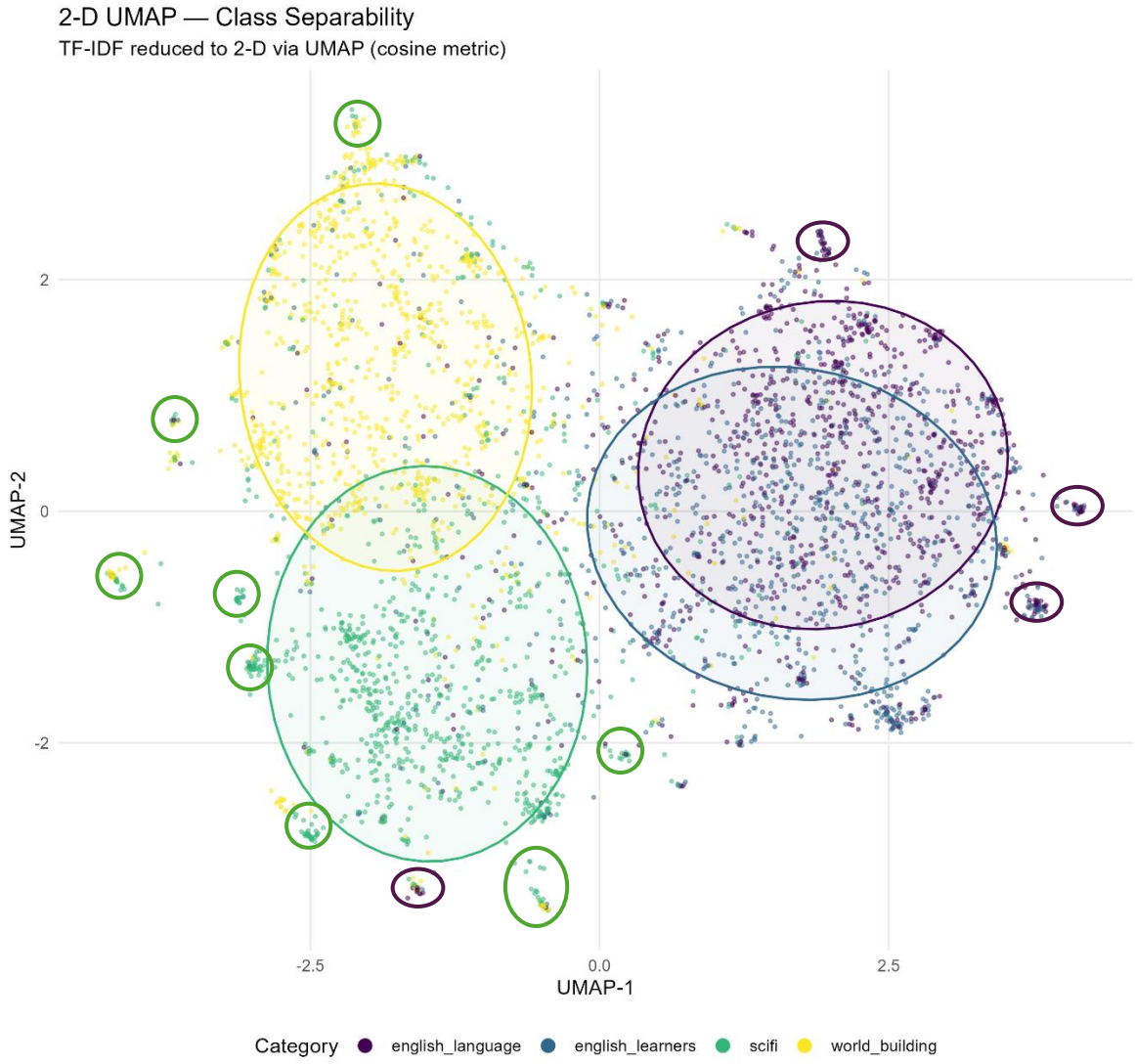


Figure 8

This can be visualised using a 2D UMAP projection of class separability (McInnes, Healy & Melville, 2018), where ELR and ELU have significant overlap. Uniquely to ELU, purple dots appear in Sci-Fi and World-Building, supporting the greater TTR and

MATTR ELU has over ELR, along with more frequent out-of-circle clusters highlighted in purple, smaller circles.

World Building and Sci-Fi also share overlap, though at a substantially smaller magnitude. Furthermore, World Building shows the greatest number of out-of-circle clusters highlighted in greener, smaller circles, suggesting this class has the highest lexical diversity, supporting the MATTR finding in (3.2 Complexity and Length.

With this analysis, it can be concluded that the confusion matrix is likely to be dominated by ELR and ELU errors, and smaller WB and Sci-Fi errors, whereas off-diagonal pairs in figure 7 will display the least.

4. Modelling and Analysis

4.1 Feature Engineering

Three feature blocks have been designed:

Table 2: Feature Blocks

Block	**Variables**	**Count**	**Rationale**
TF-IDF text	Stemmed unigrams from Title + body; stopwords removed; doc-freq >= 5; sparsity <= 0.99	1284	Lexical content (Porter, 1980; Salton & Buckley, 1988)
Stylistic Counts	n_first_person, n_second_person, n_third_person, n_modals, n_question_marks, n_exclamations, n_caps_tokens	7	ELU/ELL stance gap not resolvable by topic words (Pennebaker, Mehl & Niederhoffer, 2003)

Structural & Engagement	Score_log, Views_log, Answers_log, TextLength, TitleLength, WordCount, n_codeblocks, n_links, n_blockquotes, n_images, n_lists, views_per_answer, score_per_view, is_unanswered	14	Site-conventional formatting + post-hoc engagement signal
Total		1305	

These were then combined into a dgCMatrix.

4.1.1 TF-IDF features

TF-IDF features consist of the body and title concatenated, stop-words (‘the’, ‘and’, ‘is’, etc) filtered. They utilise porter stemming (Porter, 1980) where similar words are reduced to their root form, for example, ‘connected’, ‘connection’, and ‘connecting’ are stemmed from ‘connect’, thus they all become the root.

Terms appearing fewer than five times are filtered as they add noise without meaningful signal, such as accidental or idiosyncratic tokens. This prevents the model from overfitting to words without empirical support. The threshold of five was chosen as a conservative rule balancing matrix stability with lexical coverage. Lastly, terms appearing in fewer than 1% of posts are filtered for the same reason, resulting in 1284 terms.

4.1.2 Stylistic Counts

This includes first/second/third person pronouns, question and exclamation marks, and modal verbs like ‘could’, ‘might’, and ‘would’. These features were made to capture differences between ELR and ELU that TF-IDF alone cannot capture (Pennebaker, Mehl & Niederhoffer, 2003). Additionally, capitalised tokens are captured as it can signal emphasis (“VERY important”), formal style, title-style writing and proper nouns.

4.1.3 HTML and engagement features

Along with the raw features, derived ratios being is_unanswered (where AnswerCount == 0) to track unanswered posts, score per 1000 views, and views per answer were made for further insight. Views, Answers, and Score have also been log-transformed, the latter being signed-log to keep negative scores.

HTML-structural markers were also tracked, primarily for differentiating ELR and ELU, n_codeblocks was made to track quoted syntax or structural analysis that is more likely discussed in ELU. However, as the .csv export does not retain HTML elements, most of these variables are zero or near-zero.

All numeric features are standardised.

4.2 Train/Test split

A stratified 80/20 split was used on the classes using caret::createDataPartition.

For tuned penalised models (Lasso, Ridge, Elastic Net), 10-fold CV on the training set was implemented via glmnet::cv.glmnet, 10-fold CV is the text-classification standard (Hvitfeldt & Silge, 2021). Tree-based models (LightGBM, XGBoost), an inner 15% validation split was formed from the training set for early stopping (patience=30). Final models were refit on the training set (nrounds=best_iter). The test set is never used for model selection. Numeric features are standardised only on the training set.

95% bootstrap confidence interval was used to quantify sampling variability, reported on the best-model accuracy (Efron & Tibshirani, 1993).

4.3 Model List

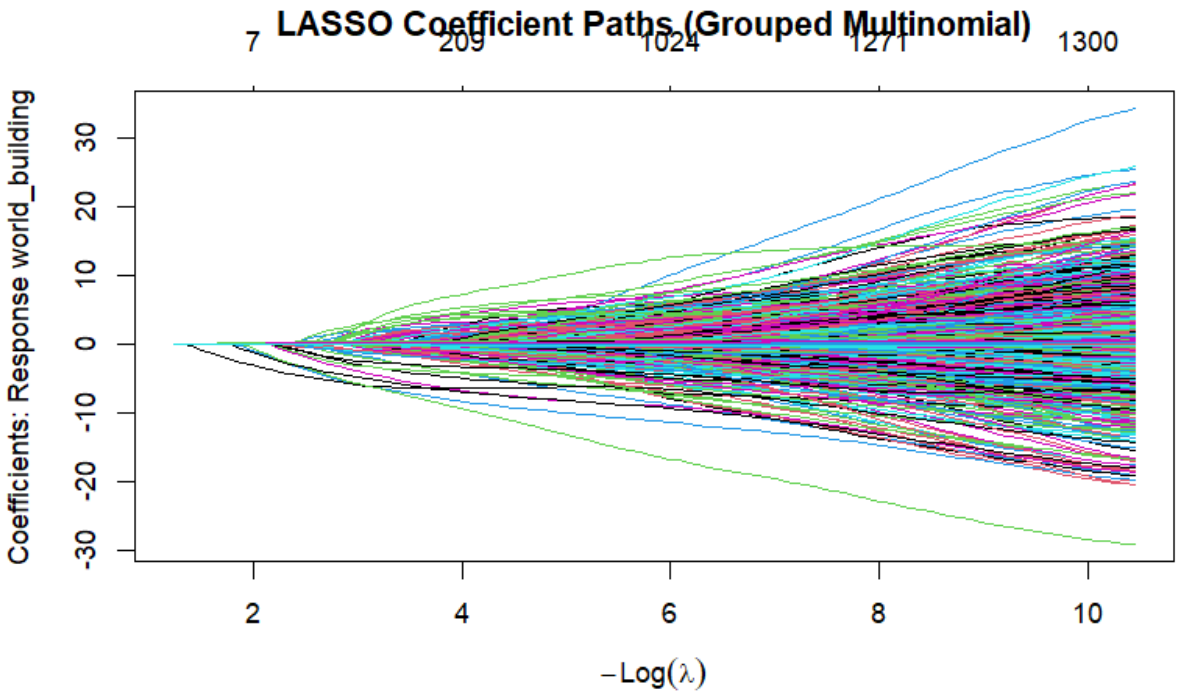
Table 3: Model List

Type	Model	Justification
Logistic Regression (LR)	Multinomial LR	Standard benchmark, it is a LIBLINEAR LR (Fan, Chang, Hsieh, Wang & Lin, 2008).
Penalised	Lasso ($\alpha = 1$)	Interpretable and sparse baseline for high-dimensional text (Tibshirani, 1996)
Penalised	Ridge ($\alpha = 0$)	A comparator for no feature selection. (Hoerl & Kennard, 1970)
Penalised	Elastic Net ($\alpha = 0.5$)	A balanced compromise between stability and sparsity (Zou & Hastie, 2005)
Naïve	Naïve Bayes (Bernoulli)	Benchmark (Wang & Manning, 2012)
Kernel	SVM (radial) on MiniLM	Non-linear similarity in dense embedding space detection (Cortes & Vapnik, 1995)
Tree-ensemble	XGBoost on sparse TF-IDF	Its histogram split finder can handle 1,000+ sparse text features natively, and excels in categorical data (Chen & Guestrin, 2016)
Tree-ensemble	LightGBM on sparse TF-IDF	Utilises exclusive feature bundling and leaf-wise growth, it excels in sparse high-dimensional data (Ke et al., 2017).
Shallow NN	fastText supervised	Utilises a bag-of-bigrams and sub-word embedding, excelling in sparse high-dimensional data (Joulin et al., 2017)
Dense NN	Average pool embedding and MLPClassifier	Dense neural network baseline, averaging tokens to a fixed-length document vector. Connected layers then learn non-linear class boundaries beyond linear TF-IDF models.

CNN	1-D CNN on token sequences	Convolutional filters applied on ordered token sequences, these capture local phrase-level patterns, which can capture unique attributes in questions (Kim, 2014).
RNN	LSTM on token sequences	Processes tokens in order which retains information from earlier words and capture long-term dependencies in questions (Hochreiter & Schmidhuber, 1997).
Ensemble	Soft voting	equal-weight probability averaging ensemble (Dietterich, 2000)

4.4 Tuning

Logistic variants' λ was selected by lambda.min in 10-fold CV.



Keras models (CNN, Dense NN, RNN, BiLSTM, LSTM) select Adam, learning rate as 1e-3, and early stopping patience was set to 4 on validation loss.

Table 4

Type	**Hyperparameter Tuning?**	**Hyperparameters**
Multinomial LR	Yes	Cost - c(0.1, 1, 10, 100) via 3-fold internal CV.
Lasso ($\alpha = 1$)	Lambda Only	Fixed alpha, lambda tuned through cv.glmnet.
Ridge ($\alpha = 0$)	Lambda Only	Same as Lasso.
Elastic Net ($\alpha = 0.5$)	Lambda Only	Same as Lasso.
Tuned Elastic Net	Yes	Alpha tuned over (0, 0.25, 0.5, 0.75, 1).
Naïve Bayes	No	-
Tuned Naïve Bayes	Yes	laplace tuned over (0, 0.5, 1, 2).
SVM (radial)	No	Baseline with cost set to 1.
Tuned SVM (radial)	Yes	uses cost x gamma grid. Cost = 2^c(-1, 1, 3) and gamma = 2^c(-7, -5, -3). On 5-fold cv.
XGBoost	Partial	nrounds is selected by early stopping on inner validation, then refit on the training data. Fixed params are max_depth = 8, eta = 0.05, subsample = 0.85, colsample_by_tree = 0.7, min_child_weight = 3.
Tuned XGBoost	Yes	Random search over params mentioned in row above, as well as gamma. Sampling 20 configs, using 3-fold cv and early stopping.
LightGBM	Partial	nrounds are selected the same way as baseline XGBoost. Fixed params are

		learning_rate = 0.05, num_leaves = 63, feature_fraction = 0.8, bagging_fraction = 0.8, bagging_freq = 5, min_data_in_leaf = 20.
Tuned LightGBM	Yes	Same random search as XGBoost.
fastText supervised	No	Fixed arguments: loss = “softmax”, learning_rate = 0.5, word_vec_size = 100, epoch = 25, max_len_ngram = 2, min_count = 1.
Dense NN	No	Fixed architecture, using early stopping and LR reduction callbacks. Avoided due to computational runtime constraints.
1-D CNN on token sequences	No	same as Dense NN.
LSTM on token sequences	No	same as Dense NN.
BiLSTM	Yes	units, dropout, embed_dim, and learning_rate are tuned via 6 configurations. It falls back to untuned BiLSTM if configurations fail.
Soft voting	Not direct	No optimised weight search due to computational runtime constraints, however it prefers tuned models where available.

All random seeds were set to student ID (34074759).

4.5 Results

The soft-voting ensemble achieves the highest performance across all metrics.

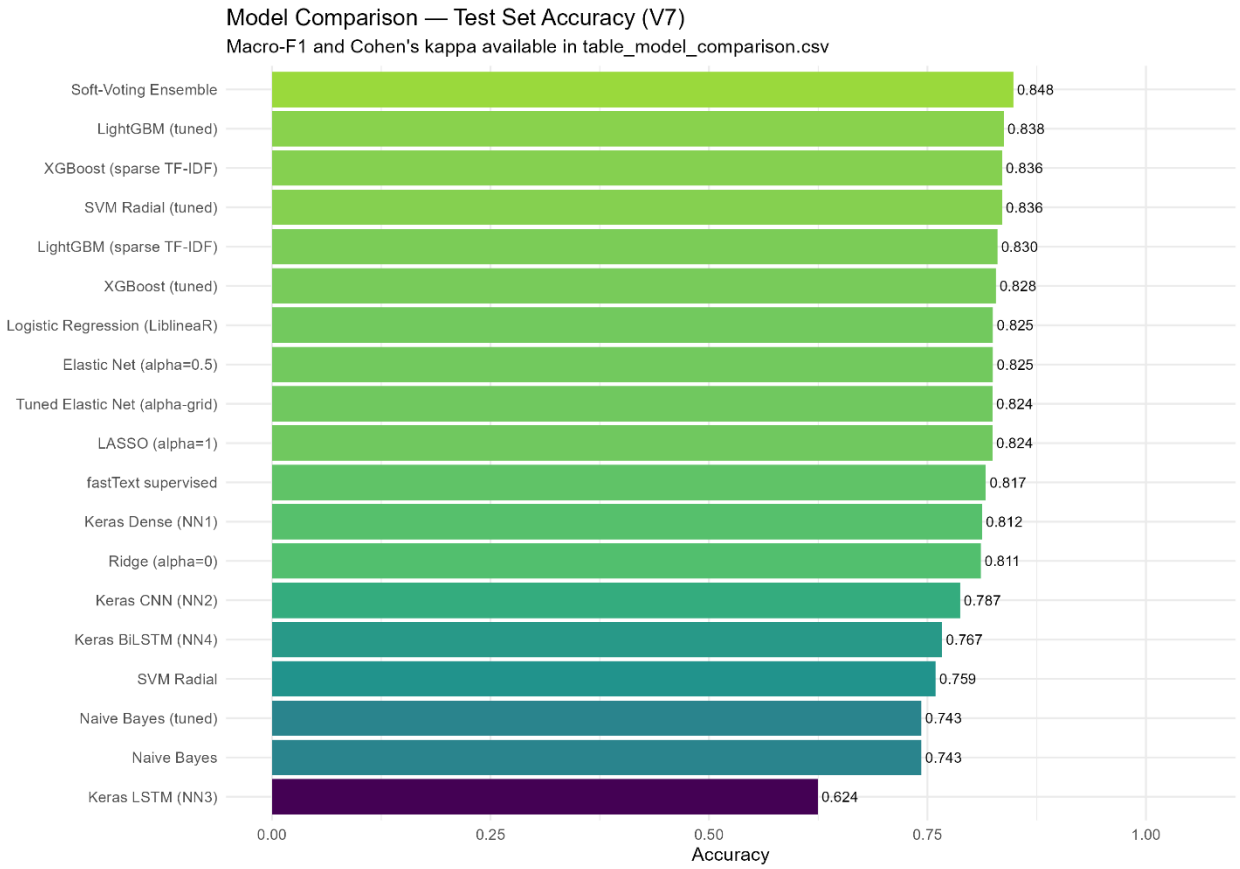


Figure 9

The best performing individual model is LightGBM (tuned), followed by XGBoost (untuned) and SVM (tuned). This is consistent with findings that histogram-based gradient boosters can give near-identical predictions (Florek & Zagdanski, 2023).

LR and penalised variants are highly competitive, with the strongest being Elastic Net and LR tied, aligning with the finding ‘linear-classifier baseline is hard to beat’ (Lin, Chen & Lin, 2023).

LSTM performs poorly compared to even the Naïve Bayes Baseline, suggesting it performs barely above chance predictions.

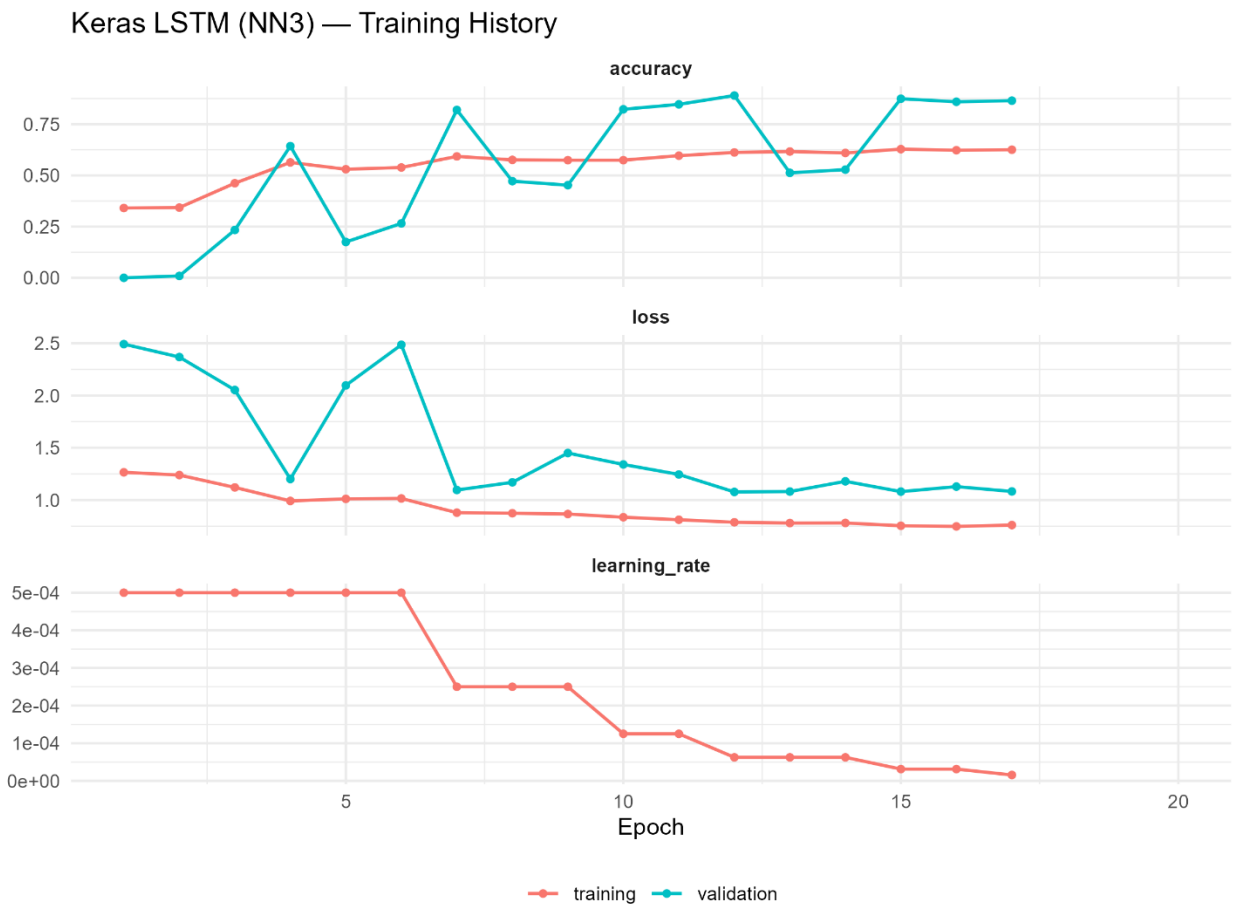


Figure 10

When looking at training history, the validation loss plateaus while the training loss converges. The corpus is too small for the task to outperform bag-of-n-grams baselines (Ezen-Can, 2020).

fastText offers competitive performance as a specialist model, however it is still outperformed by the same models as the dense NN.

Table 5: All Evaluation Metrics for Models

Model	Accuracy	Kappa	F1	Precision	Recall	Balanced Accuracy
Soft-Voting Ensemble	0.848	0.7933	0.8449	0.8451	0.845	0.8967
LightGBM (tuned)	0.8375	0.7833	0.8379	0.8388	0.8375	0.8917
SVM Radial (tuned)	0.8356	0.7808	0.8356	0.8357	0.8356	0.8904
XGBoost (sparse TF-IDF)	0.8356	0.7808	0.8361	0.8369	0.8356	0.8904
LightGBM (sparse TF-IDF)	0.83	0.7733	0.8304	0.831	0.83	0.8867
Logistic Regression (LiblineaR)	0.825	0.7667	0.8239	0.8241	0.825	0.8833
Elastic Net (alpha=0.5)	0.825	0.7667	0.825	0.8254	0.825	0.8833
LASSO (alpha=1)	0.8244	0.7658	0.8242	0.8248	0.8244	0.8829
Tuned Elastic Net (alpha-grid)	0.8244	0.7658	0.8242	0.8248	0.8244	0.8829
fastText supervised	0.8169	0.7558	0.8164	0.816	0.8169	0.8779
Keras Dense (NN1)	0.8125	0.75	0.8141	0.8179	0.8125	0.875
Ridge (alpha=0)	0.8112	0.7483	0.8107	0.8104	0.8112	0.8742
Keras CNN (NN2)	0.7875	0.7167	0.7872	0.7945	0.7875	0.8583
Keras BiLSTM (NN4)	0.7669	0.6892	0.7644	0.7752	0.7669	0.8446
SVM Radial	0.7594	0.6792	0.7574	0.7563	0.7594	0.8396
Naive Bayes	0.7431	0.6575	0.739	0.7441	0.7431	0.8288
Naive Bayes (tuned)	0.7431	0.6575	0.739	0.7441	0.7431	0.8288
Keras LSTM (NN3)	0.6244	0.4992	0.594	0.6292	0.6244	0.7496

The Macro-F1 ordering matches accuracy, which confirms the ranking of models in accuracy is not an artefact of balanced classes. This is largely the case with all other metrics.

Laplace smoothing did not improve the Naïve Bayes' performance.

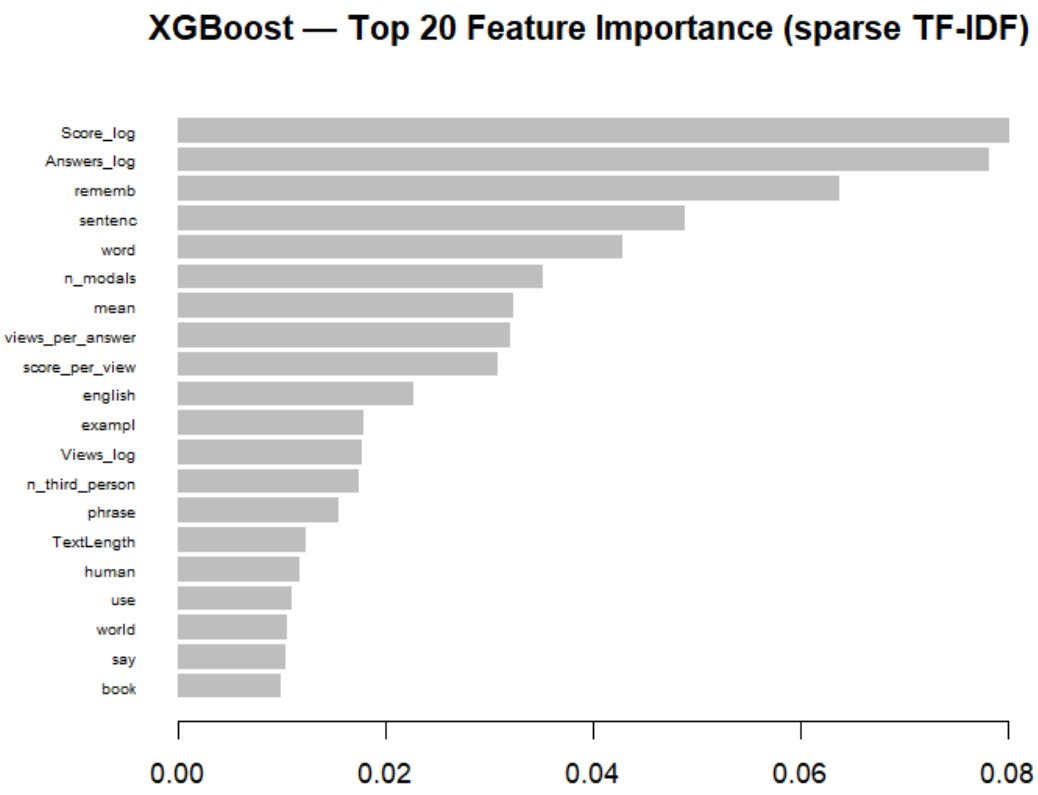


Figure 11

Log-transformed score and answers have the highest feature importance, suggesting community engagement are highly predictive of class labels, this implies systematic differences in how posts are perceived in each site. However, it also suggests the models rely on post-outcome data. This means model accuracy may not hold when a user is creating a question, and the model cannot use this post-outcome data to inform them of the correct site.

4.6 Soft-voting Ensemble

Table 6: All Metrics for Soft-Voting Ensemble

Axis	Metric	Value
Point	Accuracy	0.848
	Cohen's K (Kappa)	0.797
	Macro-F1	0.845
	Macro-Precision	0.845
	Macro-Recall	0.845
	Macro-Balanced-Accuracy	0.897
Per Class F1	Sci-Fi	0.961
	World Building	0.965
	English Language	0.728
	English Learners	0.738
Probabilistic	Multi-Class ROC-AUC (Hand-Till)	0.967
Uncertainty	95% Bootstrap CI	[0.830, 0.864]
	Bootstrap SE	0.009
Significance	McNemar χ^2vs LightGBM (Tuned)	2.49, p = 0.115
	Disagreements (b,c)	(60, 43)

The McNemar test suggests the ensemble does not significantly perform better than LightGBM (tuned), suggesting LightGBM can offer comparable performance, and this extends to SVM (tuned) and XGBoost (untuned).

The F1 for both ELU and ELR are lower than Sci-Fi and World Building, supporting implications from the EDA.

The Hand-Till ROC-AUC suggests the model has strong class separability with sparse mis-predictions, mostly between ELR and ELU.

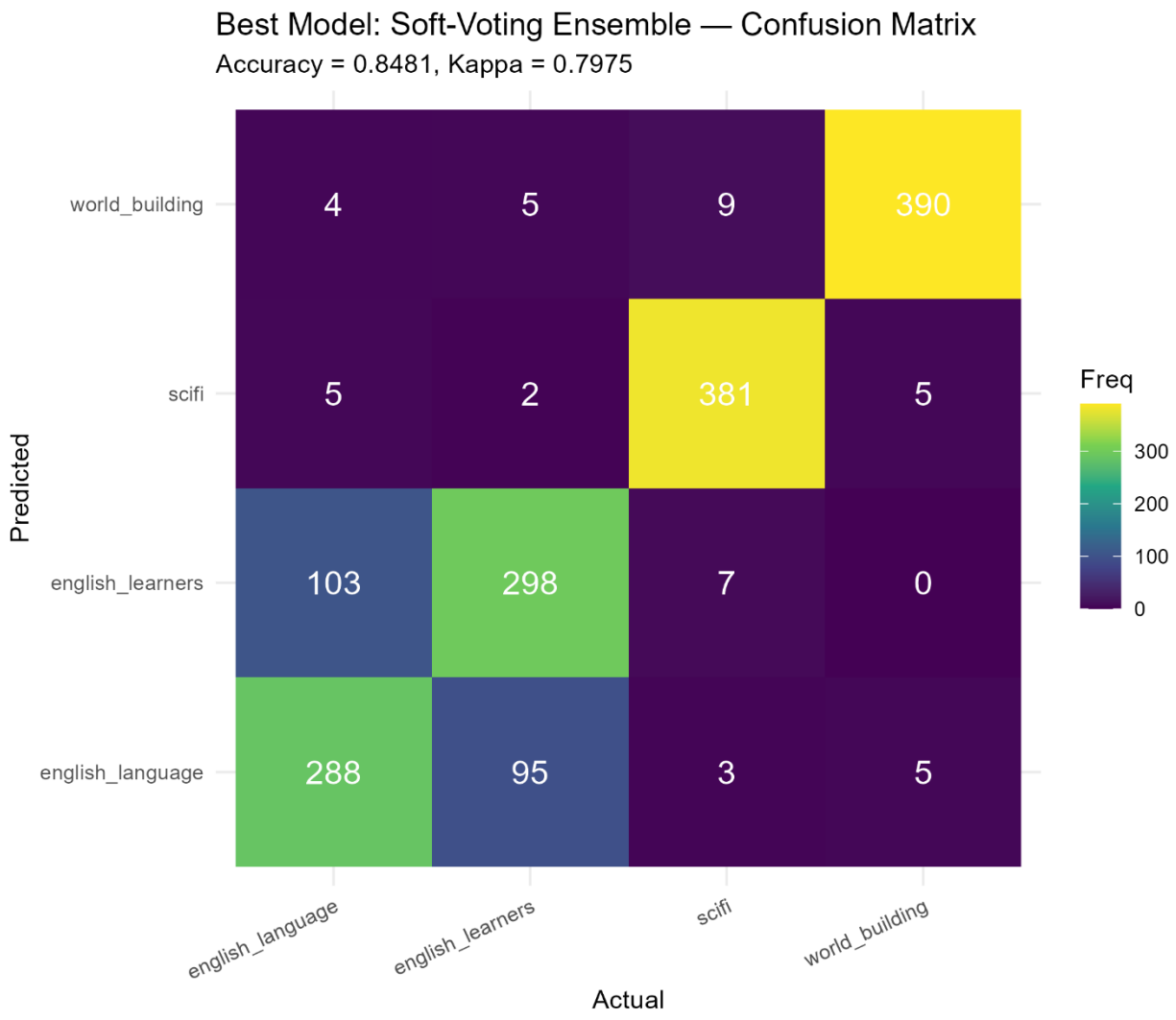


Figure 12

With the accuracy and kappa of this model, it can be concluded this model can be reliable for prediction of the correct site.

World Building contains zero confusion ELR, and very little with Sci-Fi and ELU, it is a similar case for Sci-Fi. This suggests they have a clear thematic language, with near-perfect separation from the model.

ELU is the weakest class, most often misclassified with ELR as predicted in the EDA. This is due to the lexical overlap, and the differences likely being more stylistic rather than lexical.

5. Evaluation

5.1 Interpretations

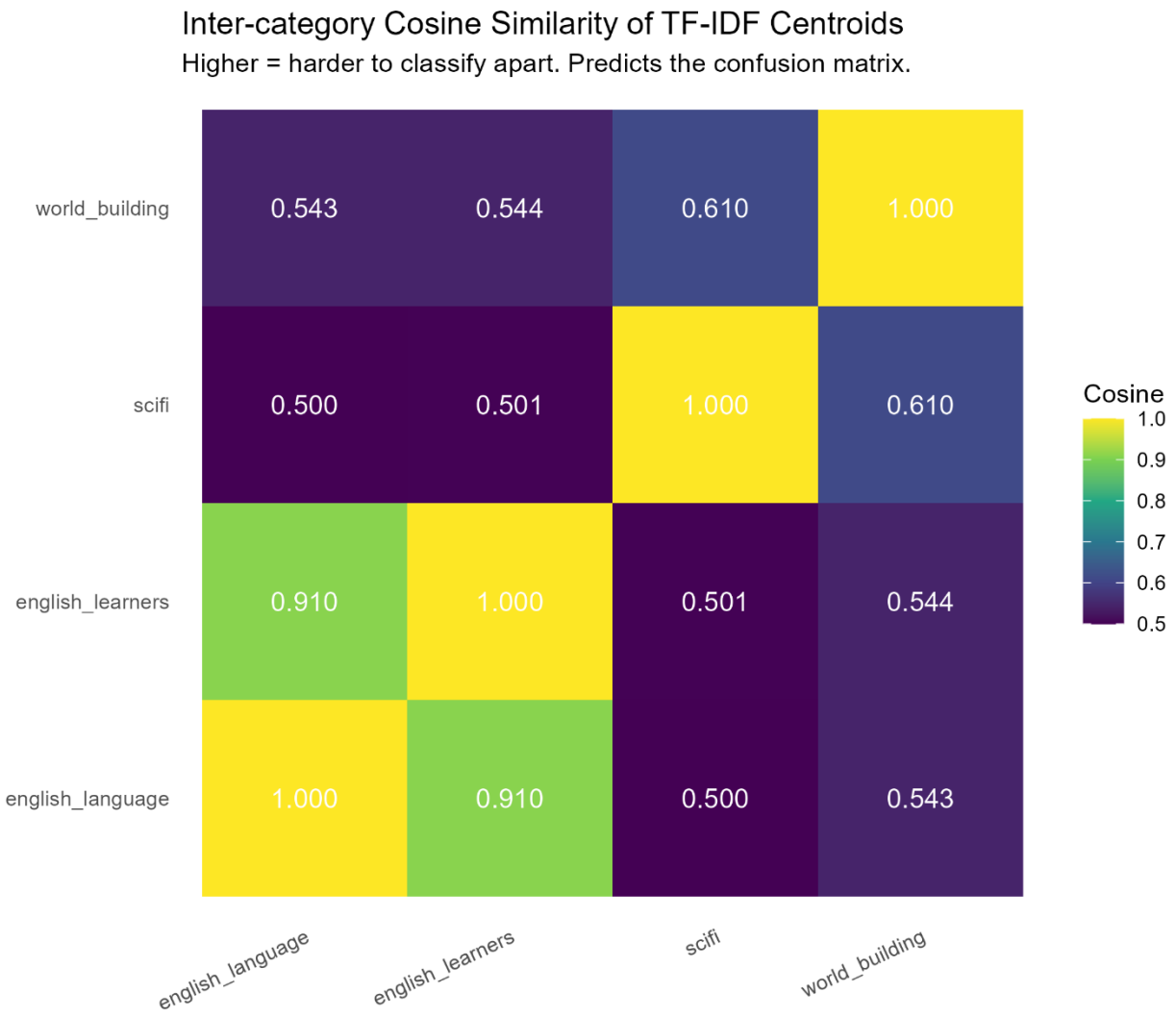


Figure 13

When looking at the cosine similarity of centroids for classes, this aligns almost perfectly with (Figure 12). This implies the model makes the exact errors the unsupervised topology predicted, suggesting the model fits real signal instead of spurious correlation while also strengthening the EDA as a diagnosis of the study.

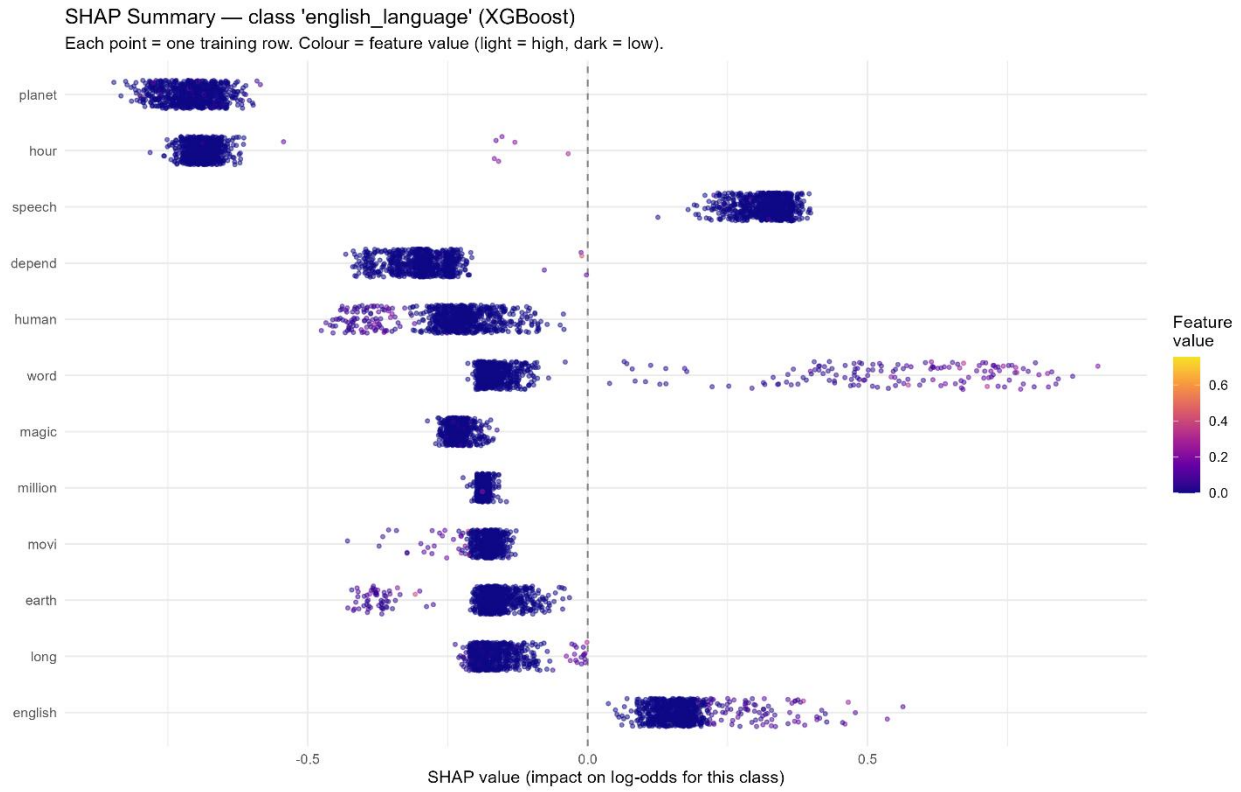


Figure 14

When considering ELU, the hardest to predict, the strongest negative drivers are ‘planet’ and ‘hour’. This means the absence of these words are exploited by XGBoost and LightGBM, pulling them towards ELU as a prediction. This aligns with observations made in (3.3 Lexical Fingerprint)

‘word’ is the strongest driver, however, it’s properties are unique. Most training rows sit at a negative SHAP value (~-0.2), while it has a sparse cluster of strong feature values at between ~0.375 and ~0.8, which pulls tree models towards ELU. This suggests its rare occurrences are decisive, but its absence suggests other sites (Lundberg & Lee, 2017)

5.2 Strengths

This study is methodologically broad with 14 models and 5 paradigms. The engineered features have also proven to be highly predictive of residual errors as seen in (Figure 11). The script is also highly reproducible with a unique seed, logged session

information, and sparse matrices ensuring training is memory-efficient. The script also has many leakage control measures; all logged in the session information file.

5.3 Limitations

Bigram TF-IDF features were not utilised in this study, where Wang and Manning (2012) found that they could close prediction gaps seen in sites sharing similarities like ELU and ELR. The Stack Exchange CC-BY-SA dataset also has biased answer-rates between sites that may not hold to the same degree in other forums like Quora and Reddit, requiring further retraining. TF-IDF vocabulary was fit on the full corpus unsupervised, while this does not leak labels, the test set still informs the feature space, in practice this impact is negligible (Hvitfeldt & Silge, 2021). Due to computational constraints, hyperparameter searches were modest, with 20 random configurations for XGBoost and LightGBM, a 3x3 cost-gamma grid for SVM and 6 BiLSTM configuration. A Bergstra and Bengio (2012) random search could have provided a more accurate prediction of classes.

6. Conclusion

This study constructs a site suggestion classifier following ethical considerations discussed in (2. Privacy and Ethical Considerations). The soft-voting ensemble provides accurate predictions for informing users of the correct site after posting a question, though it may not be as accurate for real-time rerouting. Its errors observed in (Figure 12) match the errors predicted in the EDA, validating the methodology for the study, while also providing a starting point for future research into a real-time re-routing classifier.

7. Generative AI Use

The report for this study was completely self-written. The code for this study was also self-written; however, Claude Sonnet was used for debugging and refinement of code.

For debugging, Claude assisted with four issues.

- Sparse S4 subsetting
    - Claude helped diagnose an issue in R regarding a sparse-matrix error where an S4 dgCMatrix object failed in inner train/validation subsetting.
- Probability-column alignment
    - Claude pointed out an issue I could not initially notice, where probability matrices from different models were aligned to the same class order before ensembling.
- Train/Test scaling
    - During the coding of the study, there was an issue with train/test scaling and claude suggested numeric scaling on training rows only, where the train-derived means and standard deviations were applied on test rows.
- Plot title overlap
    - For the lasso coefficient and lasso cross-validation curve plots, claude assisted me with configuring the margins to allow the title to not overlap with the top boundary of the plots.

However, the use of Claude came with mistakes that I spotted. Claude's suggestion for the plot title fix initially provided me with font sizes that became illegibly small, thus I resized them for clarity.

Claude also helped me understand why principal component analysis can be adverse for tree models on sparse TF-IDF data. Tree models make axis-aligned splits on features, whereas PCA replaces sparse word features with linear combinations. This ends up removing word-level sparsity, and splits become less interpretable and do not reflect the data.

To conclude, all final decisions on outputs and code were checked and edited by me.

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Appendix

EDA Appendix

Descriptive Statistics

Category	N	Proportion	Mean_Score	Median_Score	SD_Score	Mean_Views	Median_Views
english_language	2000	0.25	2.24	1	4.34	868	263.5
english_learners	2000	0.25	1.04	0	2.6	473	151
scifi	2000	0.25	7.87	6	6.48	1054	437

world_building	2000	0.25	6.02	4	6.86	1283	450
	Mean_Answers	**Pct_Unanswered**	**Mean_WordCount**	**Mean_TextLen**	**Mean_QMarks**	**Mean_FirstPers**	**Mean_CodeBlk**
	1.86	17.2	167	966	1.89	6.03	0
	1.7	7.8	141	790	2.1	6.28	0
	0.98	26.5	193	1070	1.04	5.18	0
	4.41	1	275	1611	1.86	5.12	0

Log-Odds Figures

Category	word	n	log_odds_weighted
english_language	noun	462	22.49386884
english_language	word	1663	19.11540876
english_language	idiom	144	12.948245
english_language	english	1102	12.58143139
english_language	comma	128	11.84718947
english_language	adjectives	117	11.52645217
english_language	phrase	702	11.44961926
english_language	grammatically	114	11.13163775
english_language	clause	279	11.09157861
english_language	term	525	11.07509345
english_language	sentence	1195	11.05504514
english_language	nouns	104	10.69995728
english_language	plural	207	10.59121168
english_language	adverb	98	10.2241973
english_language	merriam	89	9.956067023
english_language	grammar	211	9.912536735
english_language	webster	85	9.776235413
english_language	clauses	83	9.380686091

english_language	etymology	70	9.290136919
english_language	words	556	9.290132165
english_language	preposition	80	9.153377304
english_language	verbs	171	9.113181205
english_language	pronunciation	134	8.997863545
english_language	zorado	38	8.772516351
english_language	oxford	72	8.686025417
english_language	example	866	8.574166108
english_language	infinitive	68	8.545487298
english_language	verb	519	8.536191289
english_language	idiomatic	65	8.291960215
english_language	oed	55	8.225979622
english_learners	noun	325	20.07320588
english_learners	sentence	1267	15.37783596
english_learners	correct	860	13.60124207
english_learners	mean	798	12.41167683
english_learners	say	960	12.00571837
english_learners	clause	255	11.67793547
english_learners	preposition	108	11.67334873
english_learners	meaning	780	11.50088411
english_learners	past	655	11.48319138
english_learners	oxford	96	11.00226455
english_learners	adverb	97	10.99284452
english_learners	tenses	86	10.74097651
english_learners	clauses	90	10.60574765
english_learners	grammatically	87	10.38076189
english_learners	comma	87	10.3791822
english_learners	use	1108	9.832291103
english_learners	present	430	9.56791093
english_learners	idiomatic	73	9.557899421

english_learners	perfect	336	9.534282118
english_learners	nouns	67	9.109882272
english_learners	grammar	160	8.961635517
english_learners	sentences	369	8.918485716
english_learners	countable	60	8.800546113
english_learners	infinitive	61	8.704215004
english_learners	understand	442	8.65776292
english_learners	presenter	33	8.515719367
english_learners	example	712	8.513295413
english_learners	participle	58	8.492220091
english_learners	cambridge	118	8.349476066
english_learners	difference	343	8.209759695
scifi	remember	1827	22.5741901
scifi	story	1730	17.82383582
scifi	manhwa	147	16.75290244
scifi	book	1448	16.23809496
scifi	protagonist	344	14.34579733
scifi	character	879	13.22864383
scifi	read	987	12.66616482
scifi	girl	618	12.29326668
scifi	anthology	129	12.13092659
scifi	aliens	310	11.97775506
scifi	dumbledore	70	11.55959495
scifi	series	576	11.41754182
scifi	manga	192	10.88433639
scifi	spider	102	10.48622821
scifi	movie	543	10.45792022
scifi	boy	441	10.42665735
scifi	novel	468	10.41632166
scifi	main	730	10.28270542

scifi	wizard	97	10.26617293
scifi	manhua	55	10.24631207
scifi	90s	156	9.807445357
scifi	deevi	45	9.268033248
scifi	voldemort	45	9.268033248
scifi	episode	312	9.105511802
scifi	recall	317	9.04147908
scifi	scene	373	9.037363382
scifi	iirc	71	8.96990753
scifi	isekai	42	8.953740409
scifi	short	605	8.94076757
scifi	duke	40	8.737936169
world_building	civilization	225	13.80529987
world_building	merfolk	75	11.00297077
world_building	magnetic	129	10.63446717
world_building	anatomically	70	10.62982351
world_building	enchantment	69	10.55361192
world_building	density	119	10.29837107
world_building	mermaids	109	9.872991139
world_building	currents	58	9.675769751
world_building	radius	104	9.642866978
world_building	orbit	230	9.642765718
world_building	energy	701	9.493743445
world_building	tidally	55	9.422182172
world_building	planet	1077	9.315867647
world_building	asteroid	107	9.254136049
world_building	oxygen	215	9.248942965
world_building	atmospheric	95	9.214188708
world_building	worldbuilding	51	9.073052105
world_building	species	669	8.908462692

world_building	mana	95	8.793218386
world_building	warfare	85	8.667665744
world_building	zone	93	8.658821559
world_building	thermal	46	8.616778691
world_building	airships	43	8.33103338
world_building	realistic	162	8.086593171
world_building	dragons	201	8.036296976
world_building	earth	1104	8.023126056
world_building	planets	204	7.951543289
world_building	damarian	39	7.934053328
world_building	world	1440	7.865874586
world_building	feasibility	38	7.831665884